

Watcher v2 — Build Roadmap

From a message filer to a receptionist. Every item scored for urgency and ease. — 14 August 2026 — v1.14

The whole gap in one line: the pipeline can listen and file, but it cannot reply. Three outcomes exist and all three mean “put this somewhere”. Adding a fourth — **answer the customer** — is the project.

Where we stand today

Built and tested	Missing
259 passing tests, no DB or network needed	Knowledge — zero tables, zero rows
98 Python files, 13 modules	Live availability — no read path to any PMS
18 DB tables + 3 migrations	Proof of identity — matching is not verifying
Every external system behind a swappable seam	
Eval runner + CI accuracy gate, 50 golden cases, 88% baseline	
Receptionist vocabulary: 19 intents, data not code	
Unified intent taxonomy — classifier = vocabulary	
Prompt v2 — domain-specific, en/ar/mixed coverage	
Channel-neutral envelope + WhatsApp/voice adapters	
Task state machine and the autonomy gate	
Reply path wired — RECEPTIONIST_REPLY is the fourth outcome	
Persistence — conversations, turns, tasks in Postgres	
Channel-neutral core — KNOWN_LEAKS = {}, boundary test enforces it	
Python 3.13	

Totals: ~5.5 engineering days remaining. Track 0 is done, Track 1 is done, Track 2 is done except 2.4 (knowledge base). The critical path is now **2.4 (knowledge base)** and **3.1 (PMS read port)**: no facts means no answers, and no read path means no pricing, no availability and no door codes.

Track 0 — Today. Hours, not days. [COMPLETE]

All five items done.

#	Work item	Urgency	Ease	Days	Status
0.1	Set default branch to main	NOW	Trivial	1 min	DONE
0.2	Remove client name from golden set and fixtures	NOW	Easy	0.25	DONE
0.3	Decide the receptionist intent vocabulary	NOW	Easy	1 hr	DONE — 19 intents, 84 examples, 5 languages, 38 tests
0.4	Merge the eval branch	NOW	Trivial	0.25	DONE
0.5	Delete the stale nifty-johnson branch	LOW	Trivial	1 min	DONE

Track 1 — Foundations. Week 1. [COMPLETE]

All three items done.

#	Work item	Urgency	Ease	Days	Status
1.1	Stop the core speaking WhatsApp	NOW	Moderate	1.5	DONE — wa_message_id to external_id, wa_chat_id to thread_id, sender_wa_name to sender_display_name. Added channel field. KNOWN_LEAKS = {}. Alembic migration 002. MetaSettings moved to channels/whatsapp.py. to_inbound_turn() bridge added. CHANNEL_REGISTRY exempts legitimate data values.
1.2	Port the four kept scaffold files	HIGH	Moderate	1.0	DONE (prior session)
1.3	Python 3.12 to 3.13	LOW	Easy	0.5	DONE — 7 version pins updated across pyproject.toml, packages, and CI.

Track 2 — Make it a receptionist. Week 2. [2.4 REMAINS]

The product's reply path is wired end-to-end. Taxonomy is unified, prompt is v2, golden set covers all 19 intents. What remains is the knowledge base.

#	Work item	Urgency	Ease	Days	Status
2.1	Conversations, tasks and slot filling	HIGH	Hard	2.0	DONE — 7 new tables via Alembic migration 003: contacts, conversations, turns, task_rows, understandings, corrections, usage_events. ORM models in db/models.py. ConversationRepository with find_or_create_conversation, record_turn, get_active_task, save_task, create_task. task_from_row/task_to_row converters. Verification-codes table deferred to 2.4.
2.2	The reply path	HIGH	Moderate	1.5	DONE — Tool registry (TakeMessage, HandoffToHuman) validated against vocabulary terminal_tool. Receptionist function in conversations/receptionist.py: check autonomy, manage task state, return OutboundAction (ask/confirm/say/hand off). ChannelSender protocol in channels/sender.py. 8 tests.
2.3	Autonomy gate	HIGH	Easy	1.0	DONE — RECEPTIONIST_REPLY as fourth RoutingAction. Receptionist protocol in worker. After classification + identity resolution, decide_autonomy() runs before rule matching. Acting intents invoke receptionist; hand-off falls through to existing routing. ProcessOutcome carries autonomy and outbound_action. 4 new orchestration tests.
2.4	Knowledge base	HIGH	Moderate	2.0	NEXT — Facts table with sensitivity flags, prose in pgvector, and a real “I don’t know” that fetches a human. Door codes are not ordinary facts. For the demo, one property’s facts fit in the prompt — no retrieval needed.
2.5	Prompt v2 and rewrite the golden set	MED	Moderate	1.0	DONE — PROMPT_VERSION bumped to v2 with domain-specific system prompt. Golden set expanded 8→50 cases covering all 19 intents in en/ar/mixed. Recorded fixtures at 88% accuracy (6 intentional confusable-pair misclassifications). Baseline updated to 0.88. CI eval gate passes.
2.6	Intent taxonomy unification	NOW	Easy	0.5	DONE — Replaced 6-member IntentType (new_lead, existing_contact_reply, support_issue, internal_team, spam_or_noise, unclear) with 19 vocabulary-aligned intents matching intents.yaml. decide_autonomy() now recognises classified intents. Receptionist fires on real messages. All test files updated.

Track 3 — Integration and launch. Week 3.

Base360.ai is ruled out as a partner. Hostaway, Guesty and Cloudbeds all publish APIs.

#	Work item	Urgency	Ease	Days	Status
3.1	PropertySystemPort plus the first adapter — the bottleneck	MED	Moderate	2.5	PENDING — Vendor-neutral read port and FastAPI router exist (PR #14). Needs the first concrete PMS adapter. Blocks pricing, availability and door codes. <code>crm_cache</code> is the right shape; delivery is write-only, so this needs a new read port. Cache facts; never cache availability.
3.2	End to end on a real number, then measure	HIGH	Moderate	1.0	PENDING — The point at which the eval number becomes real rather than recorded.

Runs in parallel — start on day one

None of these are engineering. All of them can quietly become the reason a date slips.

#	Work item	Urgency	Ease	Days	Status
P1	Pick the first client	NOW	Easy	--	Decides which PMS adapter gets built first. Without it, 3.1 is a guess.
P2	Read the Hostaway / Guesty / Cloudbeds API docs, get sandbox keys	HIGH	Easy	0.5	Public docs, usually self-service sandboxes. Shape the port from what two or three of them offer in common.
P3	File Meta business verification	MED	Easy	1 hr	No longer blocking — unverified start is allowed. File it anyway before volume makes it bind.
P4	Graphify as a build aid (optional)	LOW	Easy	0.5	Maps the repo for the coding agents. Local parsing, no vector store, nothing leaves the machine.

The dates (revised 14 August 2026)

Milestone	Original	Revised	Status
Eval merged, name leak fixed, vocabulary decided	End of day 1	--	DONE
Core stops speaking WhatsApp; scaffold files ported	End of week 1	--	DONE
Holds a conversation and replies — rough demo	Middle of week 2	End of this week	DONE
Taxonomy unified, prompt v2, eval at 50	--	14 Aug (today)	DONE
Knowledge and safety rules in — demo-ready	End of week 2	~2 days from now	NEXT — blocked on 2.4 only (2.6 cleared)
Live availability, measured, on a real number — pilot-ready	Middle of week 3	~5 days from now	PENDING — blocked on 3.1 + P1
Phone answering	Week 4	Week 4	PENDING — speech-to-text seam already exists

What would actually move these dates

Intent taxonomy unification drifting. Cleared — 2.6 is done. This is no longer a risk.

Knowledge base scope creep. The 2.0 days assumes a structured intake form plus PMS sync. “Can it read our PDF handbook?” is a different and much larger project. Say no for now.

Quality, not features. Getting to *working* is days away. Getting to *trustworthy* — where it never invents a check-in time — is measured by the eval runner, and that number is not fully under your control. Budget a tuning tail.

PMS API access. Docs are public, but sandbox approval and rate limits are theirs to grant. Start P2 on day one; it is the new version of the Meta-verification lesson.

If you do only one thing next session: start the knowledge base (2.4). It is the last hard engineering item before a demo. The taxonomy is unified, the prompt is tuned, the reply path is wired — the receptionist just has no facts to reply with.

Session log — 14 August 2026

Session 1: PR #13 on claude/file-review-planning-dcyb2g. Tests: 228→248 (+20 new tests). Commits: 5 (one per item: 1.1, 1.3, 2.1, 2.2, 2.3) + docs. Items completed: 1.1 (1.5d), 1.3 (0.5d), 2.1 (2.0d), 2.2 (1.5d), 2.3 (1.0d) = **6.5 engineering days delivered.**

Session 2: PR #14 (OpenAPI fix, merged), PR #15 on claude/review-repo-updates-handoff-uv16t2. Tests: 248→259 (+11). Items completed: 2.5 (1.0d), 2.6 (0.5d) = **1.5 engineering days delivered.**

Cumulative: 8.0 engineering days delivered. 5.5 remaining.